



ADBMS-Expt 6

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Question 1



Experiment 06

Medium

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

$\text{Difference} = \text{Apples Sold} - \text{Oranges Sold}$

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

```
98 CREATE TABLE Sales
99 (
100     sale_date DATE,
101     fruit VARCHAR(20),
102     sold_num INT,
103     PRIMARY KEY (sale_date, fruit)
104 );
105
106 INSERT INTO Sales
107 VALUES
108 ('2020-05-01','apples',10),
109 ('2020-05-01','oranges',8),
110 ('2020-05-02','apples',15),
111 ('2020-05-02','oranges',15),
112 ('2020-05-03','apples',20),
113 ('2020-05-03','oranges',0),
114 ('2020-05-04','apples',15),
115 ('2020-05-04','oranges',16);
116
117 select sale_date,
118 sum(case when fruit='apples' then sold_num else -1*sold_num end) difference
119 from Sales
120 group by sale_date
121 order by sale_date
122
```

Data Output Messages Notifications



Showing rows: 1 to 4   Page No:

	sale_date date	difference bigint
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

Question 2



Experiment 06

Hard

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- **id** represents the current node.
- **p_id** represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

```
134
135
136 CREATE TABLE Tree
137 (
138     id INT PRIMARY KEY,
139     p_id INT
140 );
141
142 INSERT INTO Tree
143 VALUES
144 (1,NULL),
145 (2,1),
146 (3,1),
147 (4,2),
148 (5,2);
149
150
151 select t.id, (case when t.p_id is NULL then 'Root'
152     when t.id in (select distinct p_id from Tree) then 'Inner'
153     else 'Leaf' end) as type
154 from Tree t
155 |
```



	id [PK] integer 	type text 
1	1	Root
2	2	Inner
3	3	Leaf
4	4	Leaf
5	5	Leaf